To what extent can public sentiments on social media platforms such as Twitter influence national or global policies on climate change?

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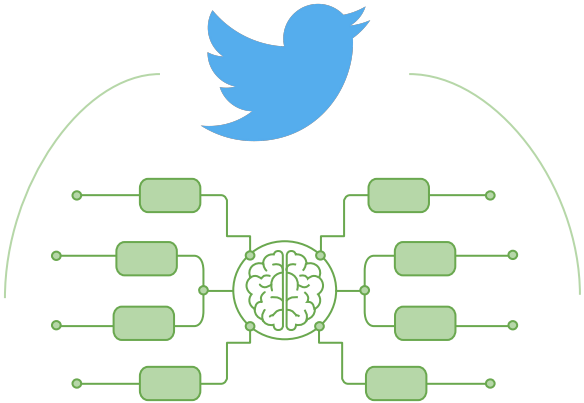
Climate Change

- Climate change refers to long-term shifts in temperatures and weather patterns.
- Since the 1800’s, The main driver of climate change is human activities, particularly the burning of fossil fuels and deforestation.

Problem

- Social media is a great resource for understanding public opinion. However, it is underutilized when policymakers have to make a decision.

- This Presentation contends that Social media analytics specifically twitter can be used to collect and analyze public sentiments on climate change, which can inform policy makers and help them make better decisions.



Solution

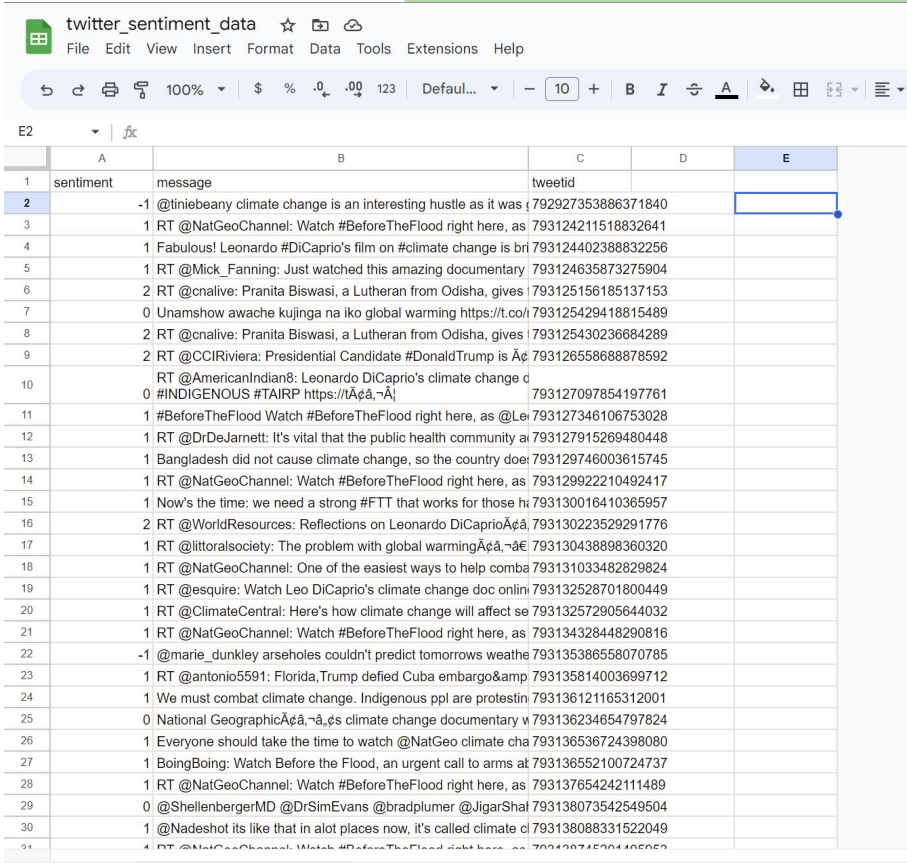
- The goal of this project is to use exploratory data analysis and logistic regression modeling to classify tweets into different sentiment categories related to climate change.

- The findings from the model can potentially inform future policy decisions that align with public preferences and contribute to preventing permanent damage to the planet.

Data Discovery

- We found twitter data from Kaggle which comprised of 43943 observations and 3 characteristics

- From here we have downloaded the data and import it as a .csv file into our program



twitter_sentiment_data

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	A	B	C	D	E
1	sentiment	message	tweetid		
2	-1	@tiniebeany climate change is an interesting hustle as it was (	792927353886371840		
3	1	RT @NatGeoChannel: Watch #BeforeTheFlood right here, as	793124211518832641		
4	1	Fabulous! Leonardo #DiCaprio's film on #climate change is bri	793124402388832256		
5	1	RT @Mick_Fanning: Just watched this amazing documentary	793124635873275904		
6	2	RT @cnalive: Pranita Biswasi, a Lutheran from Odisha, gives	793125156185137153		
7	0	Unamshow awache kujinga na iko global warming https://t.co/	793125429418815489		
8	2	RT @cnalive: Pranita Biswasi, a Lutheran from Odisha, gives	793125430236684289		
9	2	RT @CCIRiviera: Presidential Candidate #DonaldTrump is Ã¢	793126558688878592		
10	0	RT @AmericanIndian8: Leonardo DiCaprio's climate change c #INDIGENOUS #TAIRP https://tÃ¢â‚¬Â¦	793127097854197761		
11	1	#BeforeTheFlood Watch #BeforeTheFlood right here, as @Le	793127346106753028		
12	1	RT @DrDeJarnett: It's vital that the public health community a	793127915269480448		
13	1	Bangladesh did not cause climate change, so the country doe	793129746003615745		
14	1	RT @NatGeoChannel: Watch #BeforeTheFlood right here, as	793129922210492417		
15	1	Now's the time: we need a strong #FTT that works for those h	793130016410365957		
16	2	RT @WorldResources: Reflections on Leonardo DiCaprioÃ¢â	793130223529291776		
17	1	RT @littoralsociety: The problem with global warmingÃ¢â‚¬â€	793130438898360320		
18	1	RT @NatGeoChannel: One of the easiest ways to help comba	793131033482829824		
19	1	RT @esquire: Watch Leo DiCaprio's climate change doc onlin	793132528701800449		
20	1	RT @ClimateCentral: Here's how climate change will affect se	793132572905644032		
21	1	RT @NatGeoChannel: Watch #BeforeTheFlood right here, as	793134328448290816		
22	-1	@marie_dunkley arseholes couldn't predict tomorrows weathe	793135386558070785		
23	1	RT @antonio5591: Florida,Trump defied Cuba embargo&amp	793135814003699712		
24	1	We must combat climate change. Indigenous ppl are protestin	793136121165312001		
25	0	National GeographicÃ¢â‚¬â„¢s climate change documentary w	793136234654797824		
26	1	Everyone should take the time to watch @NatGeo climate cha	793136536724398080		
27	1	BoingBoing: Watch Before the Flood, an urgent call to arms at	793136552100724737		
28	1	RT @NatGeoChannel: Watch #BeforeTheFlood right here, as	793137654242111489		
29	0	@ShellenbergerMD @DrSimEvans @bradplumer @JigarShal	793138073542549504		
30	1	@Nadeshot its like that in alot places now, it's called climate c	793138088331522049		

Exploratory Data Analysis

- We have utilized ***Pandas*** to read the csv files that we have imported using “pd.read_csv”
- We have collected all the tweets and separated it all by the total count of each sentiment and the percentage
- This data then will be used to create a pie and bar chart

```
import pandas as pd
```

```
#reading data using pandas
tweets = pd.read_csv('/content/twitter_sentiment_data.csv')
tweets.head()
```

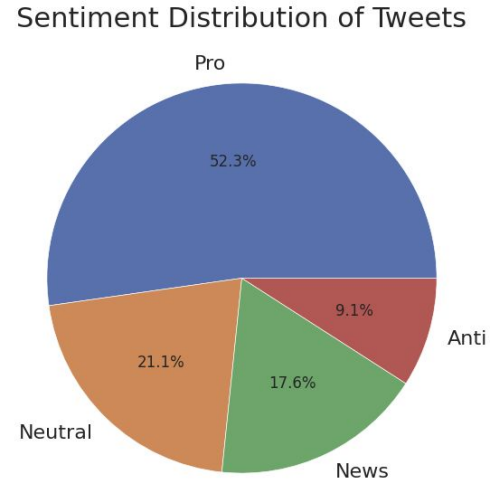
	sentiment	message	tweetid
0	-1	@tiniebeany climate change is an interesting h...	792927353886371840
1	1	RT @NatGeoChannel: Watch #BeforeTheFlood right...	793124211518832641
2	1	Fabulous! Leonardo #DiCaprio's film on #climat...	793124402388832256
3	1	RT @Mick_Fanning: Just watched this amazing do...	793124635873275904
4	2	RT @cnalive: Pranita Biswasi, a Lutheran from ...	793125156185137153

```
#get count and percentage of each category
value_counts = tweets["sentiment"].value_counts()
value_counts.name = "Total_Count"

value_norms = tweets["sentiment"].value_counts(normalize=True)
value_norms.name = "Percentage"

display(pd.concat([value_counts, value_norms], axis=1))
```

	Total_Count	Percentage
1	22962	0.522541
2	9276	0.211092
0	7715	0.175568
-1	3990	0.090799

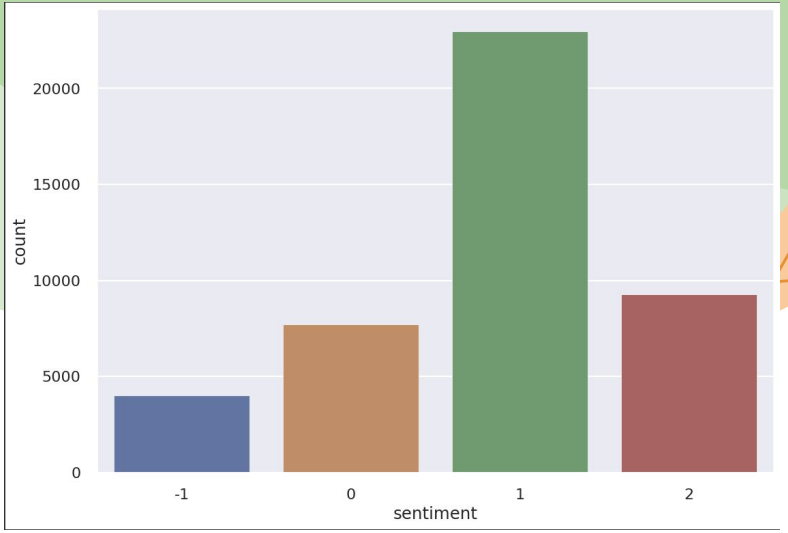
RESULTS

```
#plotting of pie chart
fig, axes = plt.subplots(figsize=(9, 7))
labels=['Pro','Neutral','News','Anti']
axes.pie(tweets["sentiment"].value_counts(),
            autopct="%3.1f%%",
            labels=labels)

plt.title("Sentiment Distribution of Tweets", fontsize=22)
plt.show()
```

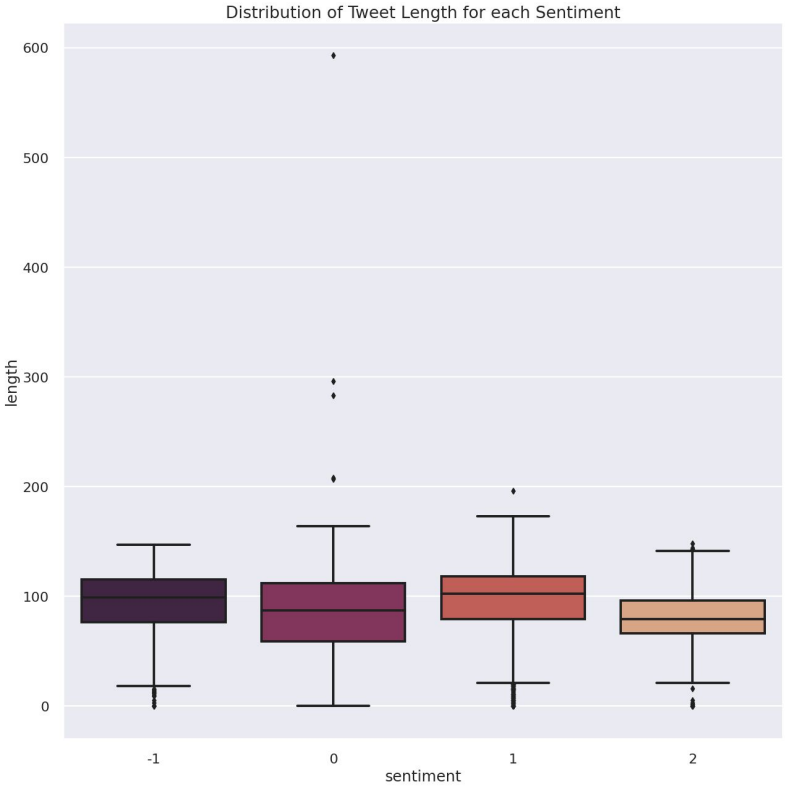


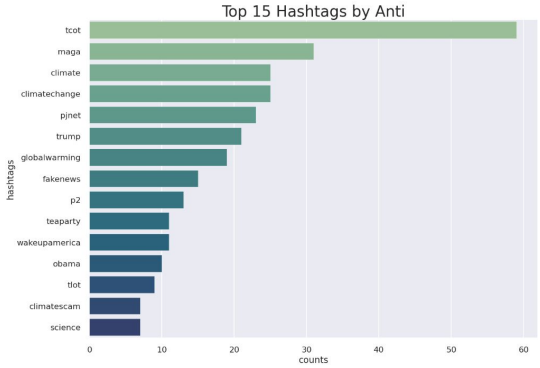

```
#Plotting of Bar Chart

sns.countplot(x ='sentiment', data = tweets)
```

```
<Axes: xlabel='sentiment', ylabel='count'>
```



RESULTS


Distribution of Tweet Length for each Sentiment
length
sentiment
-1
0
1
2
0
100
200
300
400
500
600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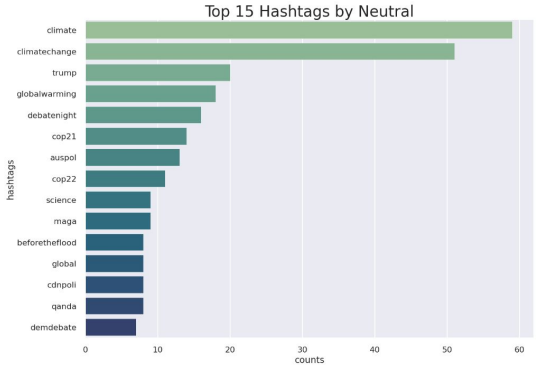
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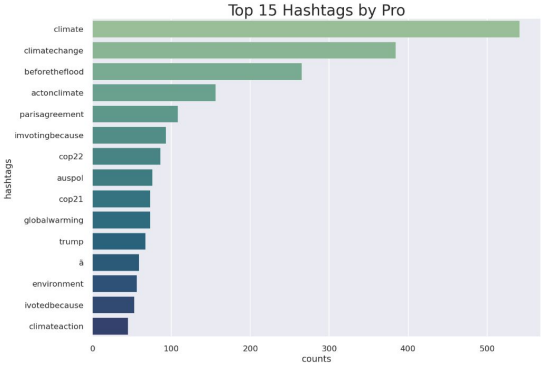
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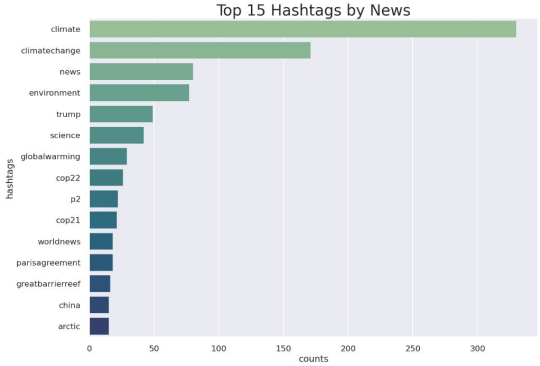
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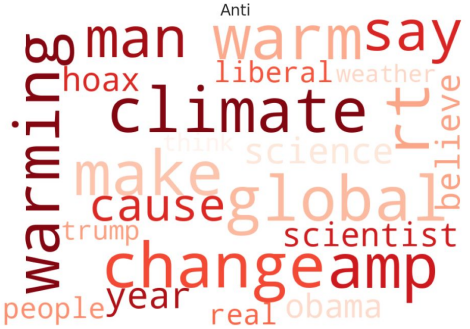
RESULTS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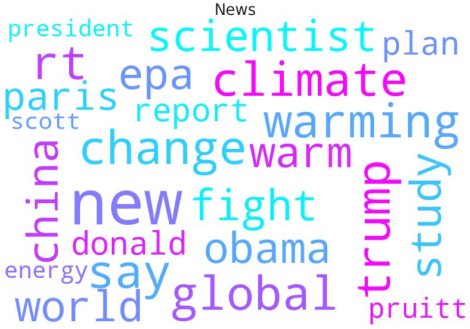
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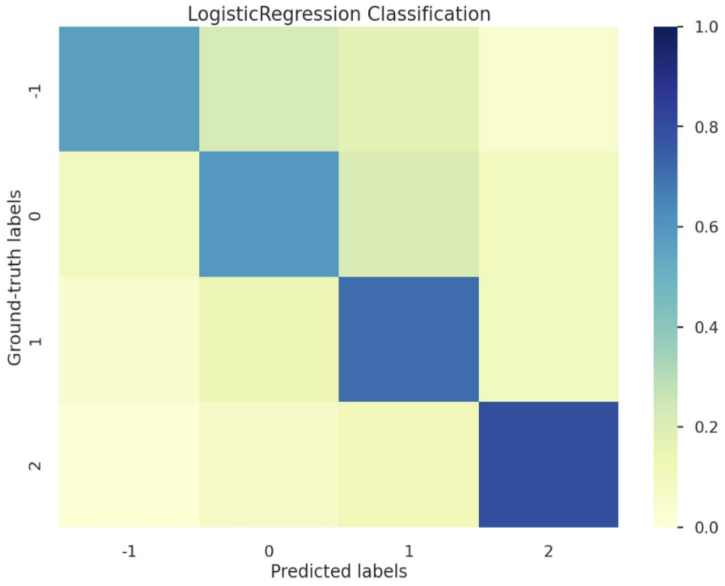
RESULTS

	precision	recall	f1-score	support
-1	0.51	0.57	0.54	633
0	0.51	0.59	0.55	1244
1	0.83	0.71	0.77	3678
2	0.69	0.79	0.74	1476
accuracy			0.70	7031
macro avg	0.63	0.67	0.65	7031
weighted avg	0.71	0.70	0.70	7031



Challenges

- The biggest challenge in this project is misinformation, especially regarding climate change denial, which has been a problem for a long time. This misinformation can come in different forms, but it still has a negative impact on research and progress towards addressing climate change.

- It is difficult to verify the accuracy of news-related tweets, which can be a major source of misinformation, as there is no way to determine the validity of the news in the first place.

- After testing five different models, the highest accuracy achieved was only 70%.

The most novel / interesting part

- During our research into social media analytics, we discovered a method called **“Crude Entity Extraction"**

- This method involves extracting information on a broad range of entities, including non-governmental organizations (NGOs), specific groups, regions, and individuals

- In the context of our project, we used Crude Entity Extraction to identify groups and individuals who promote or oppose human-caused climate change awareness.

Thank You!